Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/18/2004
PI Name:	Sekerka, Robert F. Ph.D.		
Project Title:	Lattice Boltzmann Computations of Binary Diffusion in Liquids under Stochastic Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sekerka@cmu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	412-268-2362
Organization Name:	Carnegie Mellon University		
PI Address 1:	6416 Wean Hall, Department of Physics		
PI Address 2:	5000 Forbes Avenue		
PI Web Page:			
City:	Pittsburgh	**State:**	PA
Zip Code:	15213-3890	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Moore, Craig	**Contact Phone:**	256 544-7585
Contact Email:	craig.moore@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sofonea, Victor C. (Laboratory for Mumerical Simulation and Parallel Computing in Fluid Mechanics)		
Grant/Contract No.:	NNM04AA16G		
Performance Goal No.:			
Performance Goal Text:			

The objective of this research is to further develop the Lattice Boltzmann (LB) model in binary liquids as a computational tool to better understand convective diffusion in stochastic microgravity (g-jitter). We shall approach this problem by focusing on binary liquid alloy diffusion couples. We seek to estimate the accuracy of, or make corrections to, binary liquid diffusivities measured in microgravity. We hypothesize that the Lattice Boltzmann model might be superior to computational fluid dynamics (CFD) for computation of diffusion and fluid flow in binary liquids under the influence of stochastic microgravity. We make this conjecture because the LB model, which is based on a distribution function, is intrinsically statistical and therefore more compatible with stochastic forces. The binary liquid diffusion couple will serve as a prototype problem for other experiments that have been or will be conducted in microgravity, for example, solidification (crystal growth) from binary melts. This research is likely to result in powerful computational methods to analyze the effects of g-jitter on various space experiments that involve convective diffusion in liquids. This

Task Description:	will require careful reformulation of the LB model for binary fluids to account for species of different mass for incompressible fluids whose density is a function of composition. We will use Finite Difference Lattice Boltzmann models, rather than Lattice Gas Lattice Boltzmann models, in order to allow the space lattice to be decoupled from the velocity discretization. Chapman-Enskog expansions will be used to identify the viscosity and the diffusivity and to understand and control their dependence on composition. The model will be tested against known solutions, including solutions for non-solenoidal flow and periodic g-jitter that we have obtained previously by perturbation theory and CFD. Then it will be used to explore thoroughly the effect of periodic g-jitter on diffusivities measured in microgravity. Stochastic g-jitter will be treated to ascertain its effect on measured diffusivities. We shall also use the LB model to explore boundary conditions, such as the no-slip condition for a binary fluid, at container walls. Further development of the LB model will also provide a strong basis for the analysis of complex fluid phenomena, such as two-phase flows that are important in space engineering systems, e.g., heat exchangers. This research will be conducted by the Principal Investigator in cooperation with Dr. Victor Sofonea, an expert in computer simulation of Lattice Boltzmann models. The objective of this research is to further develop the Lattice Boltzmann (LB) model in binary liquids as a computational tool to better understand convective diffusion in stochastic microgravity (g-jitter). We shall also use the LB model to explore boundary conditions, such as the no-slip condition for a binary fluid, at container walls. This research is likely to result in powerful computational methods to analyze the effects of g-jitter on various space experiments that involve convective diffusion in liquids. The development of diffuse reflection boundary conditions for LB models will provide another computational tool for modeling of fluid flow in rarefied atmospheres, such as those that may be encountered by a space plane, or in microelectromechanical systems (MEMS) in the transition regime to high Knudsen numbers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of the LB model will also provide a strong basis for the analysis of complex fluid phenomena, such as two-phase flows that are important in space engineering systems, e.g., heat exchangers. The development of diffuse reflection boundary conditions for LB models will provide another computational tool for modeling of fluid flow in the transition regime to high Knudsen numbers, such as occur in microelectromechanical systems (MEMS) .
Task Progress:	Work continued on this project in cooperation with Dr. Victor Sofonea from the Laboratory for Numerical Simulation and Parallel Computing in Fluid Mechanics, Center for Fundamental and Advanced Technical Research of the Romanian Academy, Timisoara. We have developed and tested realistic boundary conditions for the upwind finite difference simulation of isothermal fluid flow using a two-dimensional finite difference Lattice Boltzmann (FDLB) model. These boundary conditions incorporate "diffuse reflection" at container walls and are much more sophisticated than the oversimplified "bounce back" or "specular reflection" boundary conditions that were formerly used to solve LB models at lattice nodes located at container walls. For diffuse reflection, fluid particles that strike a wall get reflected at some random angle that is uncorrelated with their angle of incidence. Our diffuse reflection boundary conditions apply to walls located at half lattice spacing outside boundary nodes. They incorporate information about the temperature and velocity of a wall. This is done by creating a set of ghost nodes located at full lattice spacing outside the boundary nodes. Linear interpolation between the boundary nodes and the ghost nodes is used to estimate the distribution functions at the wall, which are assumed to obey a Maxwellian distribution corresponding to the temperature and velocity of the wall. The resulting equations are used to solve for the distribution functions in the ghost nodes. This information is used to update the boundary nodes at the next time step. The equilibrium distribution functions are expressed as series expensions up to second order in the fluid velocity. We used the diffuse reflection boundary conditions to compute velocity distributions in micro-channel flow in two dimensions, both for Couette flow and for Poiseuille flow. For Couette flow, one has parallel walls that move in opposite directions while remaining a fixed distance apart. The steady velocity profile varies linearly from one wall to the other. For very small Knudsen number, which is the ratio of the mean free path to the channel width, there is essentially no slip at the walls, but for higher Knudsen numbers, there is slip at the walls, and the difference between the velocity of the fluid near the walls to that of the walls, which we call the slip velocity, is found to be proportional to the Knudsen number, a result in agreement with kinetic theory by Maxwell. Similarly, results for Poiseuille flow (forced flow between fixed parallel walls) displayed parabolic profiles with slip velocities at the walls proportional to the Knudsen number. Therefore, this mesoscopic LB model captures an important feature of microscopic kinetic theory. Such results are important for modeling Micro-Electro-Mechanical-Systems (MEMS), where the Knudsen number can be large because the size of the system is small, or aerodynamics in rarified atmospheres, where the mean free path is large. A paper based on this work has been submitted to J. Computational Physics. We have also investigated the diffusivity of isothermal LB models. We use a FDLB model with two chemical components to compute diffusion in one-dimensional binary diffusion couple. FDLB models display an apparent value of the diffusivity, which is the sum of the physical diffusivity and a spurious numerical diffusivity. The value of the physical diffusivity is proportional to the relaxation time, and hence to the Knudsen number. The first order upwind difference scheme, which is stable, displays a large numerical diffusivity. For small Knudsen number (the classical regime), the numerical diffusivity can only be made small relative to the physical diffusivity by using an extremely fine grid size (typically 10,000 points in one dimension) which is very inefficient and would be prohibitive in two or three dimensions. Second order schemes, such as the central difference scheme, lead to much smaller or vanishing numerical diffusivity, but they are unstable or lead to spurious oscillations in cases where there are steep composition gradients. We have therefore explored flux limiter techniques, which are higher order but in which a "smoothness function" is used to control oscillations. According to the general methodology for the construction of TVD schemes with flux limiters, the updating rule of the distribution functions at a given timestep and lattice node is rewritten in a conservative form using two fluxes that are adjusted to produce smoothness of results. The first order upwind scheme can be recovered as a special case. Two flux limiters, known as Van Leer and Monitorized Central Difference, were considered. Both are quite efficient relative to the first order upwind difference scheme, perhaps by a factor of 10 for comparably accurate diffusivities. A paper based on this work had been submitted for publication to Physical Review E. Recently we have begun to explore non-isothermal FDLB models. These models require a much more elaborate set of discrete velocities to accurately compute the internal energy, and hence the temperature. Preliminary calculations

	display both slip velocities and temperature jumps at walls. Such models may be extremely useful for efficient mesoscopic modeling of microelectromechanical devices (MEMS), as well as for computing aerodynamic effects in rarified atmospheres. These results were presented at an international invited lecture (the Marion Smoluchowski Lecture) entitled "Lattice Boltzmann Simulations of Fluid Flow: Slip and Temperature Jumps at Walls," by Robert F. Sekerka in Warsaw, Poland, May 24, 2004.
Bibliography Type:	Description: (Last Updated: 02/21/2007)
Articles in Peer-reviewed Journals	Sekerka RF. "Similarity solutions for a binary diffusion couple with diffusivity and density dependent on composition." Progress in Materials Science, 2004 49(3-4):511-36. http://dx.doi.org/10.1016/S0079-6425(03)00033-1 , Jan-2004
Presentation	Sekerka, R.F. "Lattice Boltzmann Simulations of Fluid Flow: Slip and Temperature Jumps at Walls" Marian Smoluchowski Lecture, Warsaw, Poland May-2004